A Comparative Evaluation of Twelve Deep Learning Architectures for Multi-Class Tire Health Assessment from Field-Collected Images

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***Abstract*—Drivers routinely misjudge tire condition by eye, yet worn and damaged tires remain a preventable contributor to road crashes. We investigate whether a single photograph is sufficient to place a tire into one of five safety-relevant condition classes: Healthy, Low Tread, Zero Tread, Sidewall Damaged, and Uneven Wear. We contribute a purpose-built dataset of 414 tire images photographed on public roads, in garages, and at commercial-vehicle stands in Dhaka, Bangladesh. On a single stratified 70/15/15 split we benchmark twelve architectures under identical data conditions: one convolutional network trained from scratch, nine ImageNet-pretrained backbones spanning lightweight, deep, and transformer families, and two YOLO classification models. Transfer learning proves decisive, raising accuracy from 0.507 to 0.731 and overall F1 from 0.416 to 0.701, with EfficientNet-B0 ranking first. The top seven models, however, are separated by only three test images. Class-wise analysis and Grad-CAM saliency instead identify the binding constraint: Low Tread (mean F1 0.453) and Uneven Wear (0.519) fail across every architecture, and saliency maps show attention drifting onto background and wheel rims rather than tread. We conclude that additional labelled data for the two wear classes, not a larger backbone, is the productive next step.**

***Index Terms*—tire condition assessment, image classification, transfer learning, class imbalance, explainable AI, Grad-CAM, road safety**

I. Introduction

Tire condition is one of the few vehicle safety factors that is fully observable from outside the vehicle, and one of the most routinely neglected. Tread depth, sidewall integrity, and wear uniformity all degrade gradually, and the point at which a tire stops being serviceable is separated from the point at which it merely looks worn by a few millimetres of rubber. That margin is difficult to judge by eye, and the consequences of misjudging it—reduced wet braking, loss of grip, sudden deflation—appear only under the conditions in which they matter most.

The problem is more acute in contexts where formal periodic inspection is limited and a large share of road traffic consists of commercial and para-transit vehicles operating on retreaded or heavily worn tires. In such settings the realistic inspection instrument is not a tread-depth gauge in a service bay but a camera phone in the hands of a driver, a fleet supervisor, or a roadside enforcement officer. This motivates a specific technical question: *given one ordinary photograph of a tire, taken in uncontrolled lighting with a consumer camera, how reliably can a deep network assign a safety-relevant condition label?*

A. Problem Statement

We frame tire health assessment as a five-way single-label image classification problem over the classes *Healthy*, *Low Tread*, *Zero Tread*, *Sidewall Damaged*, and *Uneven Wear*. This granularity is deliberately finer than the binary good/defective framing that dominates the published literature [1], because the classes map to different interventions: Zero Tread and Sidewall Damaged warrant immediate removal from service, Low Tread warrants scheduled replacement, and Uneven Wear signals a mechanical fault upstream of the tire itself, such as misalignment or incorrect inflation.

B. Motivation

Two properties of this problem make it a poor fit for the assumptions that underlie most published tire-inspection systems. First, the imagery is *in the wild*: variable illumination, arbitrary viewing angle, partial occlusion by the vehicle body, and cluttered backgrounds, rather than the controlled X-ray or factory-line acquisition used in industrial quality control [2], [3]. Second, the available data is *small and naturally imbalanced*, because healthy tires are abundant and severely degraded ones are not evenly distributed across vehicle types. Both properties are the normal condition of an applied deployment and both are under-represented in the literature, which tends to report on large curated industrial corpora.

C. Objectives and Contributions

The contributions of this work are as follows.

1) **A field-collected dataset.** We construct and document a 414-image, five-class tire condition dataset captured under real roadside conditions, together with the sampling protocol used to counteract the natural overrepresentation of well-maintained private vehicles (Section III-A).
2) **A controlled twelve-model benchmark.** We train and evaluate twelve architectures—a from-scratch CNN, nine ImageNet-pretrained backbones, and two YOLO classification models—on an identical stratified split with a shared metric implementation, so that reported differences are attributable to the model rather than to the data pipeline (Section IV).
3) **Class-wise and saliency-based error attribution.** We show that error is concentrated in two specific classes across every architecture, and use Grad-CAM [4] to attribute part of the residual error to background and wheel-rim evidence rather than tread evidence (Section IV-E).
4) **An explicit account of the data constraint.** We report the resolution limit imposed by a 67-image test set and show that differences among the leading models fall within it, so that the benchmark is read as evidence about the dataset rather than as an architecture recommendation.

II. Literature Review

A. From Handcrafted Features to End-to-End CNNs

Early automated tire inspection relied on handcrafted texture descriptors applied to radiographic images under controlled acquisition. The field has since converged on end-to-end convolutional classification. Pratiwi and Rijati [1] trained a custom CNN from scratch on 1,856 tire images to separate defective from qualified tires, comparing four train/test ratios and reporting a best accuracy of 76.67% at an 80:20 split. Two findings from that work are directly relevant here: the split ratio measurably changed reported performance, and train/validation divergence persisted across every configuration despite augmentation, which the authors attribute to limited dataset size.

B. Transfer Learning as the Dominant Strategy

Saleh *et al.* [2] fine-tuned nine ImageNet backbones—VGG16/19, ResNet50, ResNet152V2, DenseNet121, MobileNetV2, InceptionV3, InceptionResNetV2, and Xception—on tire X-ray imagery and found Xception the strongest, despite a large domain gap between natural ImageNet photographs and radiographic tire scans. This result establishes transfer learning rather than from-scratch training as the default for tire imagery, and motivates our decision to treat the from-scratch CNN as a baseline to be beaten rather than as a candidate solution.

C. Multimodal and Industrial Pipelines

More recent work increases architectural sophistication. Mignot *et al.* [3] pair Mask R-CNN instance segmentation with a separate multimodal severity classifier that fuses image features with tire metadata, producing both a defect localisation and a severity grade. The system is strong, but it presupposes resources that a roadside or fleet-level deployment does not have: large annotated corpora, controlled multi-camera acquisition, and per-zone fine-tuning. Its two-stage structure nonetheless confirms that tire condition assessment is industrially significant enough to justify substantial engineering effort.

D. Explainability in Visual Inspection

Grad-CAM [4] produces class-discriminative saliency maps by weighting convolutional feature maps with the gradients of the target class score, requiring no architectural modification and no retraining. It has become the standard instrument for asking whether a visual classifier is attending to task-relevant evidence. We adopt it diagnostically: not to confirm that a correct prediction was correctly reasoned, but to characterise *why* particular classes fail.

E. Identified Gaps and Our Position

Three gaps emerge from the reviewed literature.

Gap 1: binary framing. The prevailing formulation is defective versus qualified [1]. This collapses conditions that demand different interventions; separating Low Tread from Zero Tread from Uneven Wear is what makes the output actionable.

Gap 2: controlled acquisition. X-ray and factory-line imagery dominate [2], [3]. Performance under uncontrolled roadside capture, with natural lighting and cluttered backgrounds, is largely unreported.

Gap 3: single-architecture reporting on large corpora. Studies typically report one proposed architecture, tuned on thousands to hundreds of thousands of images. Practitioners working with a few hundred images have little evidence about which architecture family to select, or whether the choice matters at all relative to the data constraint.

This work addresses all three: a five-class formulation, field-collected uncontrolled imagery, and a twelve-model comparison under a fixed data budget.

III. Methodology

Fig. 1 summarises the complete pipeline. The design principle throughout is that every model must see exactly the same data under exactly the same conditions, so that any observed difference is attributable to the architecture.

A. Dataset

1) Sourcing and Curation: Images were captured with consumer smartphone cameras at public roads, apartment garages, police impound areas, and rickshaw/tempu stands in Dhaka, Bangladesh. Three observations during collection shaped the final protocol.

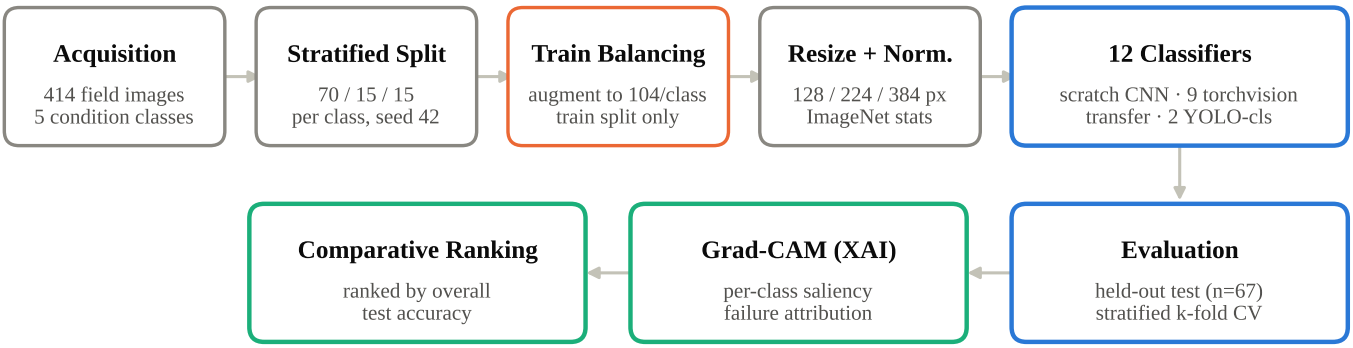


Fig. 1. End-to-end experimental pipeline. All twelve models consume the identical stratified split produced once with a fixed seed. Augmentation is applied to the training partition only; validation and test partitions remain 100% real images throughout. Evaluation, ranking, and Grad-CAM analysis share a single metric implementation so that results are directly comparable across models.

First, illumination proved to be the dominant quality constraint. An initial evening session produced a high discard rate due to underexposure, and collection was subsequently restricted to daylight hours.

Second, and more consequential for dataset validity, privately owned cars were found to be systematically well maintained, and therefore to yield almost exclusively Healthy samples. Sampling these vehicles in proportion to their road presence would have inflated the majority class without adding discriminative information. We therefore *deliberately undersampled private vehicles* and preferentially sampled commercial and para-transit vehicles—rickshaws, CNG autorickshaws, rental vehicles, buses, trucks, covered vans, and pickups—which carry a far higher incidence of low tread, worn-out tread, and sidewall damage. This is a conscious departure from representative sampling in favour of class coverage, and we note it explicitly as a property of the dataset.

Third, a structured capture protocol was adopted: for each eligible tire the vehicle type was recorded, the visible condition was assessed, and separate tread and sidewall photographs were taken. This mirrors, at much smaller scale, the zone-based acquisition philosophy used in industrial systems such as [3].

2) Statistical Breakdown: The final dataset contains 414 real images across five classes. The class distribution is naturally imbalanced, with Healthy at 149 images (36.0% of the corpus) and Sidewall Damaged at 55 (13.3%)—a majority-to-minority ratio of 2.7:1. Table I gives the full breakdown and Fig. 2 visualises it.

3) Partitioning: A stratified split was applied per class at a 70/15/15 ratio with a fixed seed of 42, yielding 287 training, 60 validation, and 67 test images. Because the per-class counts are small, integer truncation makes the realised test fraction range from 15.0% to 16.7% and the validation fraction from 14.0% to 14.8%. The split is generated once and reused by every model, and the test partition is evaluated exactly once per model, after all tuning is complete.

TABLE I
DATASET COMPOSITION BY CLASS AND PARTITION

Class	Total	Train	Val	Test	Train$^+$
Healthy	149	104	22	23	104
Zero Tread	78	54	11	13	104
Low Tread	75	52	11	12	104
Uneven Wear	57	39	8	10	104
Sidewall Damaged	55	38	8	9	104
Total	**414**	**287**	**60**	**67**	**520**

Train$^+$ denotes the training partition after augmentation-based balancing. Validation and test partitions are never augmented.

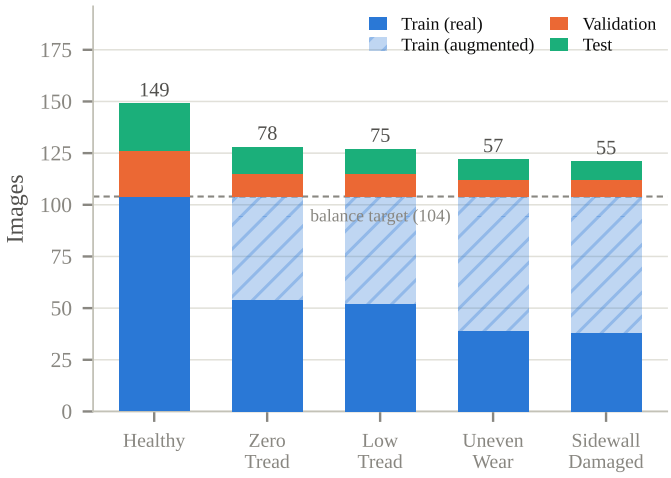


Fig. 2. Class distribution before and after balancing. Solid bars are real images; the hatched region is the synthetic supplement added to the training partition only. Every class is raised to the majority count of 104, so the minority classes rely on proportionally more synthetic material.

4) Preprocessing and Augmentation: Training images are balanced by generating synthetic copies of minority-class images until every class reaches the majority count of 104, giving a balanced training partition of 520 images. Each

synthetic copy applies one randomly selected transformation: rotation by a uniformly sampled angle in $[-15°, +15°]$, a horizontal flip, or a brightness scaling factor drawn uniformly from $[0.7, 1.3]$. Source images are drawn with replacement.

Two consequences follow, and we state them plainly because they bear on the results. The balancing is nominal rather than informational: Uneven Wear reaches 104 training images from only 39 distinct originals (1.67 synthetic copies per original) and Sidewall Damaged from 38 (1.74 per original), whereas Healthy reaches the same count from 104 distinct images. Sampling with replacement further means some originals are duplicated several times while others contribute once. The minority classes therefore carry substantially less visual variation than their nominal count suggests.

Images are then resized and normalised per model: 128×128 with tensor scaling only for the from-scratch CNN, and 224×224 or 384×384 with ImageNet channel statistics for the pretrained backbones. The YOLO classification models use their framework's native $[0, 1]$ scaling.

B. Model Architecture

1) From-Scratch Baseline: The baseline is a three-block convolutional network: $3 \to 16 \to 32 \to 64$ channels with 3×3 kernels and unit padding, each block followed by ReLU and 2×2 max pooling, then an adaptive average pool to 4×4, a 128-unit fully connected layer, dropout at $p = 0.3$, and a five-way output layer. It carries approximately 0.16M parameters and serves as the reference point against which transfer learning is measured.

2) Transfer-Learning Backbones: Nine ImageNet-pretrained backbones were selected to span the architectural design space rather than to maximise any single score: lightweight models (SqueezeNet 1.1, MobileNetV3-Large, RegNetY-400MF, EfficientNet-B0), deeper convolutional models (ResNet-18, ResNet-50, DenseNet-121), a modernised convolutional model (ConvNeXt-Small), and a pure transformer (ViT-B/16). Two YOLO classification variants, YOLO11n-cls and YOLOv8x-cls, were added at opposite ends of the capacity range.

For each backbone the ImageNet classification head is replaced by a five-way linear layer and training proceeds in two phases. Phase one freezes the backbone and trains only the new head at a higher learning rate, allowing the randomly initialised head to adapt without corrupting pretrained features through large early gradients. Phase two unfreezes the backbone and fine-tunes the full network at a learning rate one to two orders of magnitude lower. Checkpointing is on best validation accuracy throughout, so the weights carried to test evaluation are never the final-epoch weights.

3) Evaluation Protocol: All models are scored by a single shared metric implementation computing accuracy, precision, recall, and F1, both per class and overall. Models are ranked by overall test accuracy. For five of the twelve models, stratified k-fold cross-validation was additionally run on the combined real training and validation images ($n = 347$, augmented images excluded), using class-weighted loss per fold in place of re-augmentation.

TABLE II
PER-MODEL TRAINING CONFIGURATION

Model	Px	Batch	Epochs	LR (P1 / P2)
CNN (scratch)	128	32	40	10^{-3} (grid)
SqueezeNet 1.1	224	32	5 / 15	10^{-3} / 10^{-4}
ViT-B/16	224	16	5 / 15	10^{-3} / 10^{-5}
RegNetY-400MF	224	32	5 / 30	10^{-3} / 10^{-4}
MobileNetV3-L	224	16	20	10^{-3} (frozen)
DenseNet-121	384	16	15 / 25	10^{-3} / 10^{-5}
ResNet-50	384	16	10 / 40	10^{-4} cosine
ResNet-18	224	32	5 / 15	10^{-3} / 10^{-4}
YOLO11n-cls	224	32	40	framework default
ConvNeXt-S	224	16	40	10^{-4}
YOLOv8x-cls	224	16	40	framework default
EfficientNet-B0	224	32	5 / 30	10^{-3} / 10^{-4}

P1/P2 denote the frozen-head and fine-tuning phases. Models using AdamW additionally apply weight decay 10^{-2} (DenseNet-121) or 10^{-4} (ResNet-18/50, SqueezeNet, ViT) and label smoothing of 0.1.

TABLE III
TEST-SET PERFORMANCE OF ALL TWELVE MODELS, RANKED BY ACCURACY ($n = 67$)

#	Model	Accuracy	Precision	Recall	F1
1	EfficientNet-B0	**0.731**	0.730	0.702	0.701
2	YOLOv8x-cls	**0.731**	0.737	0.687	0.687
3	ConvNeXt-S	0.716	0.723	0.679	0.684
4	ResNet-18	0.701	0.702	0.658	0.667
5	YOLO11n-cls	0.701	0.704	0.657	0.667
6	DenseNet-121	0.687	0.728	0.625	0.651
7	ResNet-50	0.687	0.709	0.647	0.665
8	MobileNetV3-L	0.672	0.683	0.616	0.632
9	RegNetY-400MF	0.657	0.638	0.601	0.608
10	SqueezeNet 1.1	0.612	0.566	0.564	0.559
11	ViT-B/16	0.612	0.578	0.572	0.573
12	CNN (scratch)	0.507	0.421	0.438	0.416

Precision, recall, and F1 are the overall figures across the five classes. Class-wise values are given in Fig. 4.

IV. Experiments and Results

A. Experimental Setup

All final training runs were executed on Google Colab with a single NVIDIA Tesla T4 GPU (14.9 GB), Python 3.13, PyTorch 2.11.0 with CUDA 12.8, torchvision, and Ultralytics 8.4.163. A global seed of 42 was set for Python, PyTorch, and CUDA. Table II lists the per-model training configuration.

B. Overall Comparative Performance

Table III reports the primary result, ranked by overall test accuracy. Fig. 3 shows the same data graphically.

1) Transfer learning is the decisive factor: Every pretrained model outperforms the from-scratch CNN, and the margin is substantial: accuracy rises from 0.507 to 0.731, an improvement of 22.4 percentage points, and overall F1 from 0.416 to 0.701. This reproduces, on uncontrolled roadside imagery, the pattern that [2] reported on radiographic imagery, and confirms that the ImageNet prior transfers usefully despite the domain gap. Notably, even the weakest pretrained model

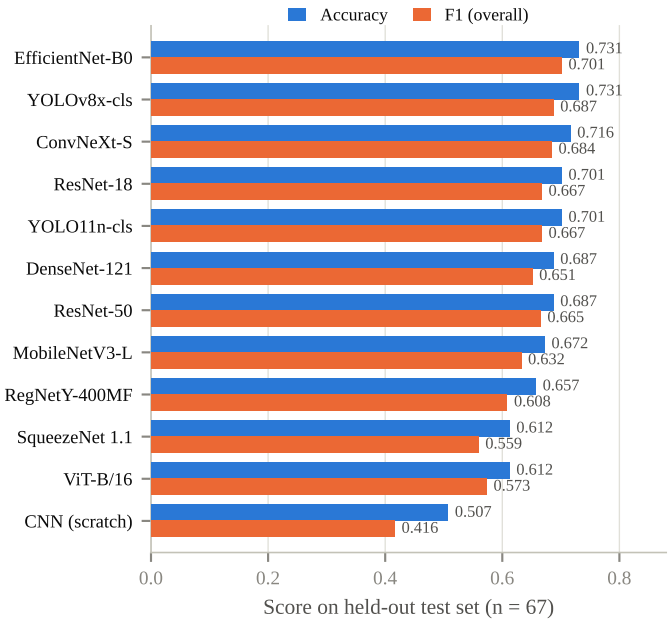


Fig. 3. Accuracy and overall F1 for all twelve models on the held-out test set. The gap between the from-scratch baseline and every transfer-learning model is large; the gaps among the top seven transfer models are not.



Fig. 4. Class-wise F1 across all twelve models. The vertical structure dominates the horizontal: class difficulty is nearly constant down every column, whereas differences between models within a column are small. Low Tread and Uneven Wear fail for every architecture tested.

(SqueezeNet 1.1, 1.2M parameters) exceeds the from-scratch CNN by 10.5 accuracy points, indicating that the benefit comes from the pretrained representation rather than from parameter count.

2) Differences among the transfer models fall within the resolution of the test set: With $n = 67$, a single misclassified image moves accuracy by 1.49 points. Converting the leaderboard to counts, ranks 1–2 classify 49 images correctly, rank 3 classifies 48, ranks 4–5 classify 47, and ranks 6–7 classify 46. The entire top seven—spanning lightweight, deep, modernised convolutional, and detection-derived architectures—is separated by three images.

We therefore do *not* claim that EfficientNet-B0 is superior to ResNet-50 on this task. The defensible claim is narrower and more useful: on a dataset of this size, architecture selection within the pretrained family is not the lever that moves performance. The same observation applies to the ordering within each tied accuracy group, where overall F1 differs by margins that a single reassigned image would erase.

C. Class-Wise Analysis

Overall accuracy conceals the structure of the error. Fig. 4 gives class-wise F1 for every model, and the column means expose a pattern that is essentially invariant to architecture.

Averaged over all twelve models, class-wise F1 is 0.826 for Healthy, 0.702 for Sidewall Damaged, 0.629 for Zero Tread, 0.519 for Uneven Wear, and 0.453 for Low Tread. The ordering is stable: no model ranks Low Tread above Healthy, and the best Low Tread F1 achieved by any architecture (0.583, EfficientNet-B0) is below the *worst* Healthy F1 achieved by any architecture (0.766, ViT-B/16). The standard deviation down the Healthy column is 0.038, against 0.121 for Low Tread and 0.137 for Uneven Wear—that is, models agree closely on the easy class and diverge only where all of them are weak.

This is the empirical core of our findings. The limiting factor is not model capacity but the representation of two specific classes, both of which are minority classes whose nominal balance was achieved through duplication rather than through additional real examples.

Two distinct mechanisms appear to be at work. Low Tread and Zero Tread both represent advanced tread degradation and differ only by degree, so the boundary between them is genuinely subtle in a photograph; the models effectively collapse a continuous physical quantity toward its endpoints. Uneven Wear is a separate failure: it is a *spatial* pattern, namely differential wear across the tread width, rather than a *magnitude*, and a single photograph taken at an arbitrary angle frequently does not show the full tread width required to observe it.

TABLE IV
STRATIFIED k-FOLD CROSS-VALIDATION ON REAL IMAGES ($n = 347$)

Model	k	**CV accuracy**	**Std. dev.**	**Test acc.**
YOLO11n-cls	3	0.718	0.023	0.701
ResNet-18	3	0.654	0.017	0.701
ViT-B/16	3	0.628	0.022	0.612
SqueezeNet 1.1	3	0.582	0.042	0.612
CNN (scratch)	5	0.444	0.060	0.507

D. Cross-Validation

Because a 67-image test set permits only weak conclusions, stratified k-fold cross-validation was run on the 347 real training and validation images for five models. Results are given in Table IV and Fig. 5.

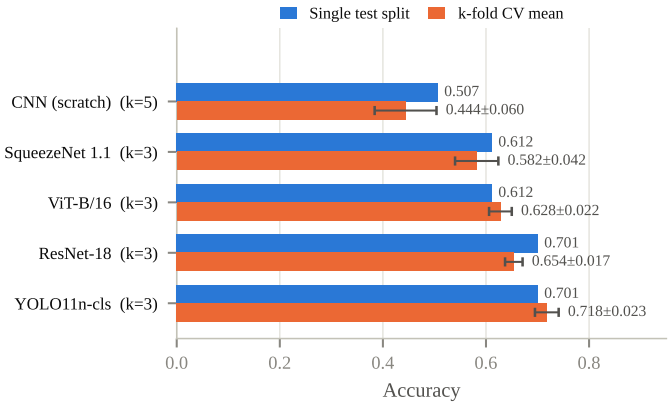


Fig. 5. Cross-validation mean accuracy (with standard deviation) against the single test-split accuracy. Cross-validated estimates track the test estimates closely, and fold-to-fold variance is smallest for the strongest models.

Three points are worth drawing out. First, the cross-validated estimates track the single-split estimates closely; the largest discrepancy is the from-scratch CNN, whose CV mean (0.444) sits well below its test accuracy (0.507), suggesting its single-split result was mildly favourable. Second, fold-to-fold standard deviation decreases monotonically with model quality, from 0.060 for the from-scratch CNN to 0.017 for ResNet-18—a stable model is stable across resamplings, which is itself evidence that the stronger models have learned something transferable rather than split-specific. Third, YOLO11n-cls achieves a higher cross-validated accuracy (0.718) than its single-split test accuracy (0.701), and on the cross-validated measure it is the strongest of the five models evaluated this way.

We note directly that cross-validation coverage is incomplete: the top three models by test score were not cross-validated, and the folds used shorter training schedules than the final runs, so the CV figures are conservative and are not directly comparable across different k.

E. Explainability Analysis

Grad-CAM [4] was applied at the final convolutional stage of five models to one test image per class. The five span the full performance range of the benchmark: EfficientNet-B0, YOLOv8x-cls, ConvNeXt-S and ResNet-18 from the leading group, and the from-scratch CNN as the baseline. Fig. 6 shows the resulting saliency maps.

Three failure modes are visible and each corroborates a quantitative finding.

Background leakage. On the Healthy sample (column 1), saliency for ConvNeXt-S and, to a lesser extent, EfficientNet-B0 extends onto the wall behind the tire rather than concentrating on the tread; the from-scratch CNN is the extreme case, placing almost all of its response on the bright vertical edge at the left of the frame. With 287 training images the models have limited opportunity to learn that background is irrelevant, and cluttered roadside capture makes background highly correlated with vehicle type, which is in turn correlated with condition—a plausible route by which a spurious shortcut becomes predictive on this dataset.

Wheel-structure confusion. The Sidewall Damaged sample (column 3) is a spoked wheel, and every model places substantial saliency on the rim and spokes. Four of the five misclassify it as Healthy; only EfficientNet-B0 recovers the correct label. Sidewall Damaged is the class whose visual evidence is most easily confounded by wheel hardware, and it is also the smallest class at 55 images.

Intermediate-state collapse. Every model misclassifies the Uneven Wear sample (column 4), and the four pretrained models all collapse it to Zero Tread with saliency correctly located on the tire but the label wrong. The models are looking in the right place and still failing, which is the signature of insufficient class-discriminative training data rather than of a localisation failure. This is the visual counterpart of the class-wise pattern in Fig. 4.

Baseline contrast. The from-scratch CNN (bottom row) produces visibly different maps from the pretrained models: instead of compact regions on the tire body its saliency traces high-contrast edges and object outlines across the whole frame, and it assigns the correct label in none of the five samples. The comparison makes the quantitative gap in Table III concrete—the pretrained backbones are not merely more accurate, they attend to different image content.

V. Conclusion and Future Works

A. Summary of Findings

We evaluated twelve deep learning architectures for five-class tire condition assessment on a purpose-built dataset of 414 field-collected images. Transfer learning was decisively better than training from scratch, raising accuracy by 22.4 percentage points and overall F1 from 0.416 to 0.701, and this held for every pretrained backbone including the smallest. EfficientNet-B0 ranked first at 0.731 accuracy, but the top seven models are separated by only three test images, so the ranking should not be read as an architecture recommendation. Class-wise analysis identified the real constraint: Low Tread (mean F1 0.453) and Uneven Wear (0.519) fail across all twelve architectures. Grad-CAM corroborated this, showing background leakage, wheel-structure confusion, and correctly localised but incorrectly labelled intermediate wear.

B. Limitations

Despite achieving a functional five-class tire-condition classification system, several limitations were identified during dataset construction, model development, and evaluation. These limitations may affect the generalizability and reliability of the reported results.

1) Limited dataset size. The dataset contains a total of 414 original tire images, which is relatively small for training and evaluating deep learning-based image classification models. Although augmentation increased the training set to 520 images, augmented images are derived from existing samples and therefore do not provide the same level of diversity as independently collected images. Furthermore, the test set

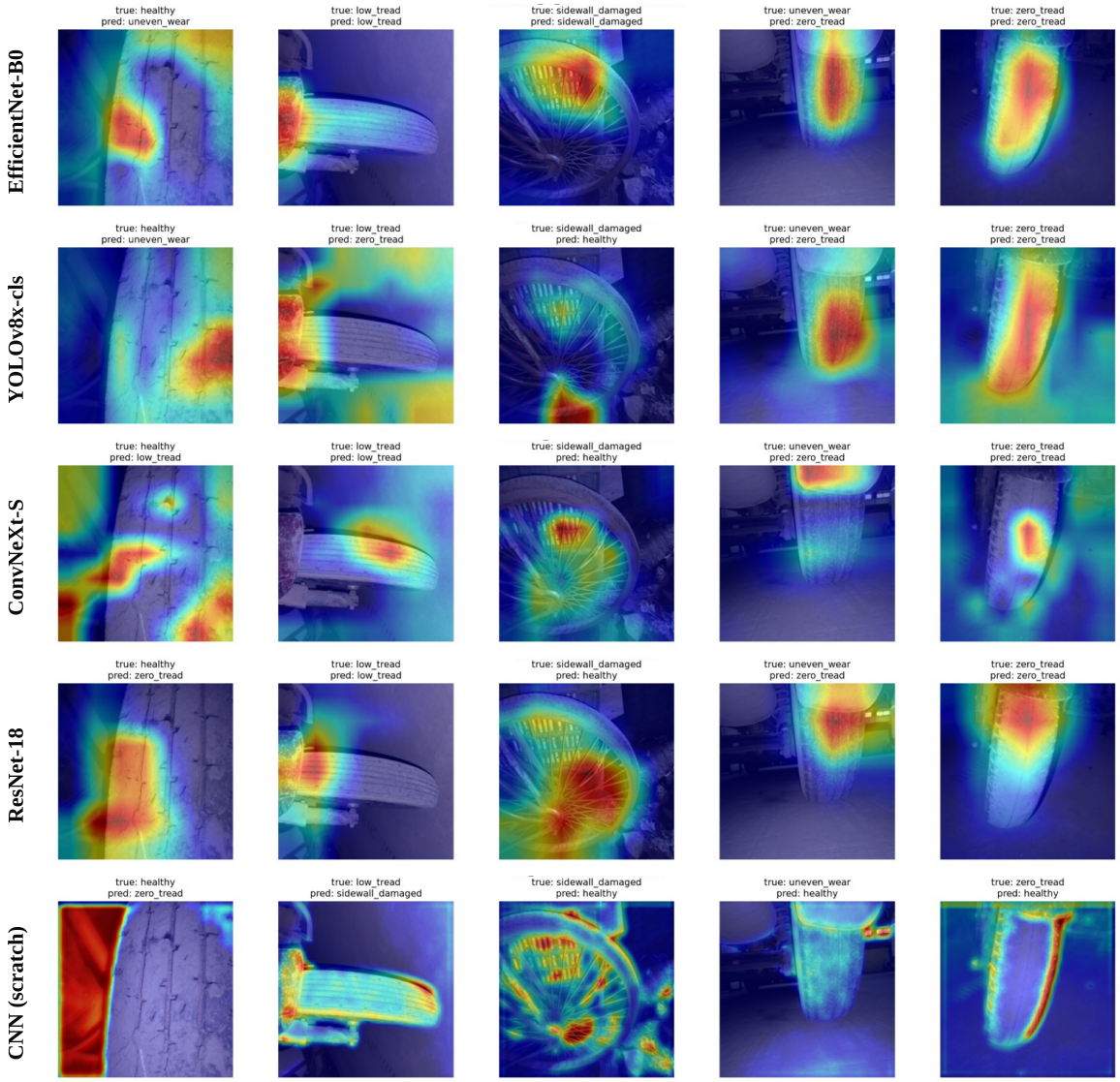


Fig. 6. Grad-CAM saliency for five models on one test image per class, in rank order with the from-scratch baseline last. Warmer regions contributed more to the predicted class. Note the attention on background wall in column 1, on rim and spokes rather than sidewall in column 3, the Uneven Wear → Zero Tread failure shared by all five in column 4, and the diffuse edge-following saliency of the from-scratch CNN in the bottom row.

contains only 67 images, meaning that each individual test prediction has a relatively large effect on the reported accuracy. Consequently, small differences in model accuracy should be interpreted with caution.

2) Native class imbalance. The original dataset does not contain an equal number of samples across the five classes. The class distribution consists of 149 Healthy, 78 Zero Tread, 75 Low Tread, 57 Uneven Wear, and 55 Sidewall Damaged images. Thus, the Healthy class contains substantially more samples than the Sidewall Damaged and Uneven Wear classes. Although augmentation and sampling were used to increase the representation of minority classes, these additional samples were generated from a limited number of original images. Therefore, the resulting dataset may still contain less visual diversity for minority classes than suggested by the final sample count.

3) Variability in image acquisition conditions. The collected images exhibit considerable variation in exposure, image resolution, lighting conditions, camera distance, and framing. Tires were photographed under different environmental and positional conditions, resulting in differences in the visibility and appearance of tread patterns and sidewall damage. Such uncontrolled variations introduce additional sources of complexity that are independent of the actual tire condition. Consequently, some models may learn image-acquisition characteristics rather than features exclusively associated with tire condition.

4) Unequal representation of tire and vehicle types. The dataset does not contain an equal representation of tires originating from different types of vehicles. In addition, the

distribution of tire conditions varies considerably across the collected vehicle types. The original class distribution itself demonstrates this imbalance, with Healthy tires accounting for 149 images, compared with 57 Uneven Wear and 55 Sidewall Damaged images. This distribution may not accurately represent the prevalence of different tire conditions across the broader population of vehicles. As a result, the model's performance may be influenced by the particular composition of vehicles and tire conditions represented in the dataset.

5) Similarity between Low Tread and Zero Tread classes. A further limitation is the visual similarity between the Low Tread and Zero Tread categories. Both classes represent severe levels of tread degradation, and the distinction between them can be subtle in an image. This creates an inherently difficult classification boundary and may result in confusion between the two categories. Based on observations during dataset development, treating Low Tread and Zero Tread as a single degraded-tread class could have produced a more clearly distinguishable classification problem. However, they were retained as separate classes to preserve the original classification objective.

6) Difficulty in collecting defective tire samples. At the beginning of the study, it was expected that defective tires would be readily available during field data collection. However, substantially fewer defective tires were encountered than anticipated. Healthy tires were considerably easier to find and collect, while tires exhibiting specific forms of severe degradation were comparatively less accessible. This created a significant class-imbalance and dataset-skew problem, particularly for the Uneven Wear and Sidewall Damaged categories. The limited availability of defective samples also restricted the diversity of examples that could be included in these classes.

7) Limited availability of uneven-wear samples. The Uneven Wear class was particularly difficult to populate because this condition is associated with several contributing factors, including improper wheel alignment, inappropriate tire pressure, hard braking, tire slipping, and other vehicle or driving-related factors. These conditions are not consistently present across vehicles, making suitable examples comparatively difficult to encounter during field data collection. Consequently, only 57 original images of Uneven Wear were obtained, making it one of the smallest classes in the dataset. The limited number and diversity of such samples may reduce the model's ability to generalize to the full range of possible uneven-wear patterns.

Overall, these limitations indicate that the reported performance should be interpreted primarily as an evaluation on the collected dataset rather than as a definitive measure of performance on all real-world tire conditions. A larger and more systematically balanced dataset, with greater diversity in vehicle types, tire conditions, imaging environments, and acquisition settings, would be required to establish stronger generalizability.

C. Future Work

Our results point to data-side rather than model-side work as the priority.

Targeted collection. The evidence indicates that additional real images of Low Tread and Uneven Wear would yield a larger return than any architectural change. On the basis of the class-wise variance, roughly doubling these two classes is the single highest-value next step.

Richer augmentation. Composing two or three transformations per synthetic copy, cycling source images rather than sampling with replacement, and adding perspective and shadow augmentation would raise effective diversity without new collection.

Background suppression. Grad-CAM evidence of background leakage motivates segmenting or cropping the tire region before classification, which would also make Uneven Wear's spatial signature easier to expose.

Standardised capture. Uneven Wear is a cross-tread pattern that a single oblique photograph often does not show. A capture protocol that guarantees full tread-width visibility, or a multi-view input, addresses this at the source.

Protocol tightening. Future iterations should cross-validate every reported model at a common k and select on validation only, so that the test partition informs no decision other than the final reported figure.

Scope extension. Manufacturing-date (DOT code) extraction was scoped out of this phase and remains a natural extension, since service age and visual condition are complementary rather than redundant safety signals.

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